Recitation 7

Bias, Standard Error, and Confidence Intervals

Review and Practice Problems

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Reading: *Probability and Statistics for Data Science*, Sections 9.2–9.3 and 9.8; Theorem 9.8, Exercise 9.3, and Example 9.45.

1 Review: bias and standard error

A parameter θ is fixed. Before observing the data, an estimator $\widetilde{g} = h(\widetilde{x}_1, \ldots, \widetilde{x}_n)$ is random; after observing $X = \{x_1, \ldots, x_n\}$, its value $g = h(x_1, \ldots, x_n)$ is a number.

Two different sources of error. For an estimator with a finite second moment,

$$\boxed{\mathrm{Bias}(\widetilde{g}) = \mathbb{E}[\widetilde{g}] - \theta, \qquad \mathrm{se}(\widetilde{g}) = \sqrt{\mathrm{Var}(\widetilde{g})}.}$$

Bias is the average signed error; standard error measures variation across repeated samples. Unbiased means $\mathbb{E}[\widetilde{g}] = \theta$, not that every estimate equals θ. Their contributions to mean squared error are

$$\mathrm{MSE}(\widetilde{g}) = \mathbb{E}[(\widetilde{g} - \theta)^2] = \mathrm{Bias}(\widetilde{g})^2 + \mathrm{se}(\widetilde{g})^2.$$

The sample mean. If $\widetilde{x}_1, \ldots, \widetilde{x}_n$ are i.i.d. with mean μ and finite variance σ^2, then

$$\widetilde{m} = \frac{1}{n}\sum_{i=1}^{n} \widetilde{x}_i, \qquad \mathbb{E}[\widetilde{m}] = \mu, \qquad \mathrm{Var}(\widetilde{m}) = \frac{\sigma^2}{n}, \qquad \mathrm{se}(\widetilde{m}) = \frac{\sigma}{\sqrt{n}}.$$

Linearity gives the mean; independence makes the cross-covariances zero when computing the variance. Multiplying n by four halves the SE.

Estimating the standard error from data. For $n \geq 2$, define

$$m(X) = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad v(X) = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - m(X))^2.$$

The sample standard deviation $\sqrt{v(X)}$ describes the spread of the *observations*. The estimated SE of their mean is $\mathrm{se}_{\mathrm{est}} = \sqrt{v(X)/n}$. Problem 1 explains the denominator $n - 1$.

2 Review: confidence intervals

A $1-\alpha$ confidence procedure produces a random interval $[\widetilde{\ell}, \widetilde{u}]$ with coverage

$$\mathbb{P}_\theta\{\widetilde{\ell} \le \theta \le \widetilde{u}\} \ge 1-\alpha$$

for each parameter value in the model. Exact equal-tail constructions may have equality; normal approximations aim for coverage near the stated level.

The normal-approximation recipe. If $\widetilde{g}$ is approximately Gaussian and centered at θ, then

$$\mathbb{P}\left(\left|\frac{\widetilde{g}-\theta}{\mathrm{se}(\widetilde{g})}\right| \le c_\alpha\right) \approx 1-\alpha, \qquad c_\alpha = F_{\widetilde{z}}^{-1}(1-\alpha/2),$$

where $\widetilde{z} \sim \mathcal{N}(0,1)$. Rearranging the event and, when needed, estimating the SE gives the observed interval

$$\boxed{\left[g - c_\alpha\,\mathrm{se}_{\mathrm{est}},\; g + c_\alpha\,\mathrm{se}_{\mathrm{est}}\right].}$$

At 95% confidence, use $c_{0.05} \approx 1.96$. The **margin of error** is the half-width $1.96\,\mathrm{se}_{\mathrm{est}}$. For an i.i.d. sample with finite positive variance, the CLT motivates

$$\left[m(X) - 1.96\sqrt{\frac{v(X)}{n}},\; m(X) + 1.96\sqrt{\frac{v(X)}{n}}\right]$$

when the sample is large enough for the normal approximation to be useful.

A population proportion. For i.i.d. $\widetilde{b}_i \sim \mathrm{Bernoulli}(\theta)$, let $\widetilde{p} = n^{-1}\sum_i \widetilde{b}_i$. Then

$$\mathbb{E}[\widetilde{p}] = \theta, \qquad \mathrm{se}(\widetilde{p}) = \sqrt{\frac{\theta(1-\theta)}{n}}.$$

From an observed proportion p, a common estimate is $\mathrm{se}_{\mathrm{est}} = \sqrt{p(1-p)/n}$. For planning before any data,

$$\theta(1-\theta) = \frac{1}{4} - \left(\theta - \frac{1}{2}\right)^2 \le \frac{1}{4} \quad \Longrightarrow \quad 95\%\text{ margin of error } \le \frac{0.98}{\sqrt{n}}$$

under the normal approximation. This worst-case choice can be conservative. The approximation needs enough expected successes and failures; it can be poor near $\theta = 0$ or 1 with small samples.

> **Interpretation.** Across repeated random samples, about 95% of intervals from a calibrated 95% procedure contain the fixed parameter. Once the interval is observed, both its endpoints and the parameter are fixed; 95% describes the procedure, not a posterior probability for the parameter.

Quick checks: A larger sample reduces SE but does not automatically remove sampling bias. Higher confidence gives a wider interval; quadrupling the sample size approximately halves its width.

Practice problems

Three core problems; Problem 4 is a backup for a fast class. The indicated times are for independent work before discussion.

Problem 1: Why divide by $n-1$? (Theorem 9.8)

Independent work: 8 minutes. Let $a_1, \dots, a_N$ be fixed population values with mean $\mu_{\text{pop}} = N^{-1}\sum_{k=1}^{N} a_k$ and variance $\sigma^2_{\text{pop}} = N^{-1}\sum_{k=1}^{N}(a_k - \mu_{\text{pop}})^2$. Draw $n \geq 2$ samples independently and uniformly with replacement, and set

$$\widetilde{m} = \frac{1}{n}\sum_{i=1}^{n} \widetilde{x}_i, \qquad \widetilde{v} = \frac{1}{n-1}\sum_{i=1}^{n}(\widetilde{x}_i - \widetilde{m})^2.$$

Prove that $\mathbb{E}[\widetilde{v}] = \sigma^2_{\text{pop}}$.

(a) Show the identity

$$\sum_{i=1}^{n}(\widetilde{x}_i - \widetilde{m})^2 = \sum_{i=1}^{n}(\widetilde{x}_i - \mu_{\text{pop}})^2 - n(\widetilde{m} - \mu_{\text{pop}})^2.$$

(b) Take expectations, using the mean and variance of $\widetilde{m}$. Where is independence needed?

(c) What is the bias if we divide the squared-deviation sum by n instead of $n-1$?

Problem 2: Participation (Exercise 9.3)

Independent work: 10 minutes. A poll records party preference and stated intention to vote:

Intention	Democrat	Republican
Certain to vote	20	30
Likely to vote	40	10
Not likely to vote	30	20

A separate study gives turnout probabilities 0.8, 0.1, and 0 for these three intention categories, respectively.

(a) Let α_R be the population fraction in the joint category "certain to vote and Republican." Assume independent, uniform sampling with replacement. Construct an unbiased estimator, derive its true standard error, and compute its observed value. Also give a numerical estimate of the SE.

(b) Estimate the Republican share *among those who actually vote*, using the turnout probabilities. State the assumptions that make this adjustment appropriate.

Problem 3: Planning the sample size (Example 9.45)

Independent work: 8 minutes. We estimate a population prevalence θ by testing n people selected independently and uniformly with replacement. Treat test results as accurate i.i.d. Bernoulli(θ) measurements. We want a 95% confidence interval with margin of error at most 0.01 (*one percentage point*). Use the normal approximation with 1.96.

(a) Without knowing θ, use $\theta(1-\theta) \leq 1/4$ to determine a sufficient sample size for this planning rule.

(b) If we plan at $\theta = 0.05$, how many tests does the rule require? Why is this smaller, and does it work for every possible prevalence?

(c) For the sample size in (b) and true $\theta = 0.05$, let $\widetilde{t}$ count the positive tests. Identify its distribution and the integer values for which $|\widetilde{t}/n - 0.05| \leq 0.01$. Write the exact probability of meeting this error tolerance as a finite sum; no numerical evaluation by hand is required.

Problem 4 (backup): A polling margin

Independent work: 5 minutes. In a random sample of 1500 voters, 55% support a proposition and 45% oppose it. Model the responses as independent Bernoulli variables with support probability θ. The population support-minus-opposition margin is $d = \theta - (1 - \theta) = 2\theta - 1$.

(a) Express the margin estimator $\widetilde{d}$ in terms of the sample support proportion $\widetilde{p}$. Is it unbiased for d?

(b) Find its true standard error and estimate that SE from the data.

(c) Construct an approximate 95% confidence interval for d. Explain why treating the support and opposition proportions as independent would give the wrong standard error.

Source: adapted from the Fall 2025 recitation source `mine/r3/r3.tex`, Section 7.9. The unbiasedness question has been added.